A1, A2, B1, B2

(3) Un sistem de calcul are un procesor ce trimite referințe

2p către memoria cache la o frecvență de 10^8 word/second, unde 10% sunt scrieri și 90% citiri.

Cache-ul are un miss rate de 10% și un block de 8 cuvinte.

BUS-ul are un bandwidth de 10^9 ~~citiri~~ accese/secundă și o lățime (bus bandwidth) de 1 cuvânt.

La orice moment, 20% din blocuri sunt dirty.

Se se calculeze procentajul din bus bandwidth folosit din cauza imperfecțiunii cache-ului în ~~fiecare~~ următoarele cazuri.

(1p) a) Cache-ul este WB + Write Allocate

A1 (1p) b) Cache-ul este WT + Write No Allocate

The rest as in A1 + nu contează procentajul de blocuri dirty

B2 (1p) a) Cache-ul este WT + Write Allocate

(1p) b) Cache-ul este WT + Write No Allocate

A2 Miss rate = 8% și un block de 16 cuvinte

20% Scrieri 80% Citiri ; Bus ~~bandwidth~~ width = 4 words

10% Blocuri dirty

(1p) a) Cache-ul este WB + Write Allocate

(1p) b) Cache-ul este WT + Write No Allocate

The rest as in A2 + nu contează procentajul de blocuri dirty

B1 (1p) a) Cache-ul este WT + Write Allocate

(1p) b) Cache-ul este WT + Write No Allocate

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③ $\boxed{A1}$

a) Accesses via the BUS $_{WB+WA}$ =

$= 10^8$ words/sec $\times 0.1 \times (0.1 \times$ Write Miss Penalty $+ 0.9 \times$ Read Miss Penalty$)$

Write Miss Penalty $= \underbrace{0.2 \times 8 \text{ BUS Writes}}_{\text{~~Allocati~~ Updati}} + \underbrace{8 \text{ BUS } \overset{\text{Reads}}{\cancel{\text{Writes}}}}_{\text{Allocati}} =$

$= 1.2 \times 8$ BUS Accesses $= 9.6$ BUS Accesses

Read Miss Penalty $= \underbrace{0.2 \times 8 \text{ BUS Writes}}_{\text{Updati}} + \underbrace{8 \text{ BUS Reads}}_{\text{Allocati}} = 1.2 \times 8 = 9.6$ Bus Accesses

$\Rightarrow$ Accesses via the Bus $_{WB+WA} = 10^7 \times 9.6$ Acc/sec.

% Used Bus $_{WB+WA} = \dfrac{10^7 \times 9.6}{10^9} = \dfrac{9.6}{10^2} = \boxed{9.6\%}$

b) Accesses via the Bus $_{WT+WNA}$ =

$= 10^8$ words/sec $\times 0.1 \times (0.1 \times$ Write Miss Penalty $+ 0.9$ Read Miss Penalty$)$

$+ 10^8$ words/sec $\times 0.9 \times 0.1 \times$ Write Hit Penalty

No write allocation, no updati!

Write Miss Penalty $= 1$ BUS Write $= 1$ BUS Acc

Read Miss Penalty $= 8$ BUS Reads $= 8$ BUS Acc.

Write Hit Penalty $= 1$ BUS Write $= 1$ BUS Acc.

Accesses via the BUS $_{WT+WNA} = 10^7 \times (0.1 \times 1 + 0.9 \times 8) + 10^7 \times 0.9 \times 1 =$

$= 10^7 (7.3 + 0.9) = 8.2 \times 10^7$

% Used BUS $_{WT+WNA} = \dfrac{8.2 \times 10^7}{10^9} = \dfrac{8.2}{100} = \boxed{8.2\%}$

③ $\boxed{B2}$

a) $\text{BUS Accesses}_{WT+WA} = 10^8 \times 0.1 \times (0.1 \times \text{Write Miss Penalty} + 0.9\ \text{Read Miss Penalty}) +$

$+ 10^8 \times 0.1 \times 0.9 \times \text{Write Hit Penalty}$

$\text{Write Miss Penalty} = \underbrace{8\ \text{BUS Reads}}_{\text{Allocati}} + 1\ \text{BUS Write} = 9\ \text{BUS Accesses}$

$\text{Read Miss Penalty} = \underbrace{8\ \text{BUS Reads}}_{\text{Allocati}} = 8\ \text{BUS Accesses}$

$\text{Write Hit Penalty} = 1\ \text{BUS Write} = 1\ \text{BUS Access}$

$\Rightarrow \text{BUS Accesses}_{WT+WA} = 10^7 \times (0.1 \times 9 + 0.9 \times 8) + 10^7 \times 0.9 \times 1$

$= 10^7 \times 8.1 + 10^7 \times 0.9 = 9 \times 10^7$

$\%\ \text{BUS Access}_{WT+WA} = \dfrac{9 \times 10^7}{10^9} = \dfrac{9}{100} = \boxed{9\%}$

b)

$\text{BUS Accesses}_{WT+WNA} = 10^8 \times 0.1 \times (0.1 \times 1 + 0.9 \times 8) + 10^7 \times 0.9 \times 1$

$= 8.2 \times 10^7$

$\%\ \text{BUS Access}_{WT+WNA} = \dfrac{8.2 \times 10^7}{10^9} = \boxed{8.2\%}$

③ $\boxed{A2}$

a) Access via the $BUS_{WB+WA} = 10^8$ words/sec $\times 0.08 \times$

$\times (0.2 \times$ Write Miss Penalty $+ 0.8 \times$ Read Miss Penalty$)$

Write Miss Penalty $= \underbrace{\frac{16 \text{ words/block}}{4 \text{ words bus width}} \text{ BUS Writes}}_{\text{Update}} + \underbrace{\frac{16}{4} \text{ Bus reads}}_{\text{Allocate}} =$

$= (0.1 \times 4 + 4)$ Bus accesses $= 1.1 \times 4 = 4.4$ Bus accesses

Read Miss Penalty = Write Miss Penalty = 4.4 BUS Accesses

BUS Accesses $_{WB+WA} = 10^7 \times 0.8 \times 4.4 = 3.52 \times 10^7$ BUS Accesses /sec

$\Rightarrow$ % BUS Used $_{WB+WA} = \frac{3.52 \times 10^7}{10^9} = \frac{3.52}{100} = \boxed{3.52\%}$

b) Accesses via the $BUS_{WT+WNA} =$

$= 10^8$ words/sec $\times 0.08 \times (0.2 \times$ Write Miss Penalty $+ 0.8$ Read Miss Penalty$) +$

$+ 10^8$ words/sec $\times 0.92 \times 0.2 \times$ Write Hit Penalty

No write allocation, no update!

Write Miss Penalty = 1 BUS Write = 1 BUS Acc

Read Miss Penalty = 4 BUS Reads = 4 BUS Acc

Write Hit Penalty = 1 BUS Write = 1 BUS Acc

Accesses via the $BUS_{WT+WNA} = 10^7 \times 0.8 \times (0.2 \times 1 + 0.8 \times 4) +$

$+ 10^7 \times 0.92 \times 2 = 10^7 \times 2.72 + 10^7 \times 1.84 = 4.56 \times 10^7$

% BUS Used $_{WT+WNA} = \frac{4.56 \times 10^7}{10^9} = \boxed{4.56\%}$

B1 (3)

a) $\text{BUS Accesses}_{WT+WA} = 10^8 \times 0.08 \times (0.2 \times \text{Write Miss Penalty} + 0.8 \text{ Read Miss Penalty}) + 10^8 \times 0.92 \times 0.2 \times \text{Write Hit Penalty}$

$\text{Write Miss Penalty} = \underbrace{4 \text{ BUS Reads}}_{\text{Allocati}} + 1 \text{ BUS Write} = 5 \text{ BUS Accesses}$

$\text{Read Miss Penalty} = \underbrace{4 \text{ BUS Reads}}_{\text{Allocati}} = 4 \text{ BUS Accesses}$

$\text{Write Hit Penalty} = 1 \text{ BUS Write} = 1 \text{ BUS Access}$

$\text{BUS Accesses}_{WT+WA} = 10^8 \times 0.08 \times (0.2 \times 5 + 0.8 \times 4) + 10^8 \times 0.08 \times 0.92 \times 1 =$

$= 10^7 \times 0.8 (0.2 \times 5 + 0.8 \times 4) + 10^7 \times 0.8 \times 0.92 =$

$= 10^7 \times 3.36 + 10^7 \times 0.736 = 4.096 \times 10^7$

$\%\text{ BUS Used}_{WT+WA} = \frac{4.096 \times 10^7}{10^9} = \boxed{4.096\%}$

b) $\text{BUS Accesses}_{WT+WNA} = 10^7 \times 0.8 \times (0.2 \times 1 + 0.8 \times 4) + 10^7 \times 0.92 \times 2 \times 1 = 4.56 \times 10^7$

$\%\text{ BUS Used}_{WT+WNA} = \frac{4.56 \times 10^7}{10^9} = \boxed{4.56\%}$